ARCHITECTURE & VALUATION REFERENCE

AI Cockpit — Enterprise Value & Governance Architecture Guide

Autonomous Agent Portfolio Valuation, Labor Capacity Intelligence & FinOps Decision Framework

System Version: 2.1.0 Enterprise Production

Reference Standard: ISO/IEC 42001 & Enterprise FinOps

Target Scale: 1,200 Enterprise Fleet / 4 Core Pilots

1. Dual-Cohort Architecture: Enterprise vs. Pilot Cohorts

The AI Cockpit provides an executive dual-cohort operational view accessible through the top navigation toggle: **[Enterprise View (1,200)]** ↔ **[Pilot View (4)]**.

Pilot View (4 Live Bots)

Built for direct operational proofs-of-concept, real bot execution loops, and real-time LLM telemetry sync.

- Interactive web chat interfaces (Ticket Triage, Copilot Studio).
- Per-employee usage tracking via live Azure OpenAI endpoints.
- Focus: Technical validation & early adoption telemetry.

Enterprise View (1,200 Agents)

Demonstrates macro-level organizational capacity unlocking, cross-functional ROI, and company-wide FinOps governance.

- 1,200 autonomous agents across 35 teams and 6 business functions.
- ~$18.2M+ annualized net value with ~83,345 hours saved/mo.
- Focus: Executive visibility, labor unlocking, and budget control.

Dimension	Pilot View (4 Agents)	Enterprise View (1,200 Agents)
Target Agents	4 Core Live Pilots (IT Support, Supply Chain, AP, HR)	1,200 Autonomous Enterprise Workflow Agents
Primary Purpose	Interactive bot execution, live token and session telemetry	Macro capacity unlocking, cross-team ROI, and FinOps budgeting
Net Value Range	~$1.2k – $3.5k / mo (pilot scale)	**~$18.2M+ Annualized** (~$1.52M / month)
Labor Capacity Unlocked	~120 – 350 labor hours / month	**~83,345 hours / month** (~520 FTEs equivalent)
Telemetry Ingestion	Direct SQLite & live Azure OpenAI completion logs	Live pilot telemetry + deterministic enterprise fleet modeling
Interactive Actions	`[Live Agent ↗]` links to live container web UIs	Fleet telemetry inspection & portfolio-wide governance

2. The 4-Tier Agent Governance Decision Matrix

Every agent in the portfolio is evaluated against four governance statuses. Rules are executed in strict sequential hierarchy: **Experiment** → **Intervene** → **Accelerate** → **Optimize**.

Status	Trigger Condition / Parameters	Business Meaning & Action Plan
🧪 **Experiment**	`Days in Production < 60 days`	**Early-Stage Ramp-Up:** Recently launched agent. Telemetry is being collected; evaluation thresholds are not penalized for low initial adoption.
🚨 **Intervene**	Triggered if **ANY** condition occurs: • `Policy Incidents > 0` (Safety alert) • `Cost vs Budget > 110%` (Overrun) • `Eval Pass Rate < 88%` (Quality drop) • `Net Value < $0` (Negative ROI)	**Immediate Risk Remediation:** Safety guardrail violation, runaway LLM token spend, or prompt degradation. System surfaces root-cause diagnostics drawer.
🚀 **Accelerate**	`Net Value ≥ $2,500/mo` **AND** `Workforce Adoption ≥ 70%` *(with Eval Pass ≥ 92% and Cost ≤ 100% budget)*	**High-ROI Scaling Candidate:** Proven top performer. Outperforms on value generation, human adoption, and accuracy. Recommended for expansion across business units.
⚙️ **Optimize**	Operates safely within standard envelope: `Eval Pass ≥ 88%`, `Cost ≤ 110%`, `Incidents = 0`, with Net Value below the $2,500/mo threshold.	**Standard Healthy Production:** Normal operational target state. Performing reliably within governance bounds. Focus is on refining prompts, latency, and token caching.

3. Mathematical Valuation & FinOps Formulas

All metrics in the AI Cockpit are derived from defensible, audited labor economics and cloud FinOps formulas:

A. Hourly Labor Rate ($/hr)

Baseline Labor Valuation

```
Hourly Labor Rate = Fully-Burdened Monthly Employee Cost / Monthly Working Hours
```

Standard enterprise default: **$6,500/mo ÷ 160 hrs = $40.63 / hour**. Configurable per department (e.g., $85.00/hr for IT Support Engineers, $95.00/hr for Cloud SREs).

B. Labor Hours Saved (Capacity Unlocked)

Productivity Impact

```
Hours Saved = Successful Executions (Eval Passed) × Baseline Human Task Duration
```

Calculates workforce capacity returned to high-value tasks. For example, 117 successful ticket resolutions × 0.25 hrs (15 min human baseline) = **29.25 labor hours saved**.

C. Gross AI Value Created ($)

Total Labor Output Value

```
Gross AI Value = Hours Saved × Hourly Labor Rate
```

The total monetary equivalent of human work executed by autonomous agents prior to deducting compute expenses. Example: 29.25 hours × $40.63/hr = **$1,188.43 Gross Value**.

D. Actual Monthly AI Cost ($)

True FinOps Cloud Spend

```
Token Cost = [(Prompt Tokens × $2.50) + (Completion Tokens × $10.00)] / 1,000,000
Actual AI Cost = Token Cost + Monthly Dedicated Infrastructure Cost
```

No artificial per-seat software tax. Only true model inference (tokens) plus dedicated container and vector compute.

E. Net AI Value Created ($)

Bottom-Line Financial ROI

```
Net AI Value = Gross AI Value − Actual Monthly AI Cost
```

The definitive bottom-line net return delivered by the agent. Example: $1,188.43 Gross Value − $70.19 AI Cost = **+$1,118.24 Net Value**.

F. Value Multiple (ROI per $1 Spent)

Capital Efficiency Ratio

```
Value Multiple = Gross AI Value / Actual AI Cost
```

Measures economic output per dollar of compute invested. At enterprise scale ($2.75M gross value ÷ $597.3k cost): **4.6× Return ($4.61 returned per $1 spent)**.

Operational Performance Formulas

Metric	Formula	Target Benchmark
Workforce Adoption	`(Active Monthly Users / Assigned Licensed Users) × 100%`	**≥ 70.0%** (Green / On Track)
Eval Pass Rate	`(Successful Sessions / Total Executed Sessions) × 100%`	**≥ 92.0%** (Warning if < 88%)
Cost vs. Budget	`(Actual Monthly AI Cost / Monthly Budget Cap) × 100%`	**≤ 100.0%** (Overrun if > 110%)

4. Enterprise Portfolio Rollup (1,200 Agents Across 35 Teams)

The enterprise fleet distribution scales across 6 core corporate functions and 35 department teams:

Business Function	Agents	Monthly Net Value	Hours Saved	FTEs Unlocked	Avg. Eval Pass
Engineering & IT	335	$822.7k	23,316 hrs	~146 FTEs	89.1%
Customer Support & Success	310	$301.8k	20,909 hrs	~131 FTEs	88.9%
Finance & Accounting	210	$245.2k	15,420 hrs	~98 FTEs	91.4%
Supply Chain & Logistics	160	$180.9k	11,280 hrs	~72 FTEs	90.2%
HR & Employee Experience	105	$95.4k	6,840 hrs	~42 FTEs	93.5%
Legal, Risk & Compliance	80	$78.1k	5,580 hrs	~31 FTEs	94.8%
Total Portfolio Fleet	**1,200**	**$1,724.1k / mo**	**83,345 hrs / mo**	**~520 FTEs**	**90.8%**

Enterprise FinOps Cost Allocation Pillars ($597.3k Total Monthly Spend)

FinOps Cost Pillar	Monthly Spend	Share	Primary Cost Drivers
Model Token Consumption	$328.1k	55%	Raw prompt & completion token inference across production LLM endpoints (GPT-4o, Sonnet).
Platform & Infrastructure	$143.2k	24%	Dedicated container app environments, GPU clusters, Kubernetes node pools, and microservice hosting.
Vector RAG & Knowledge Pipelines	$77.5k	13%	Enterprise semantic search indexing, vector embedding ingestion, and chunk re-ranking pipelines.
AI Safety, Governance & Guardrails	$47.7k	8%	PII redaction proxies, prompt injection firewalls, hallucination evaluators, and audit compliance logging.